Name: ______________________

Worksheet

Congruent triangles (SAS)

Task A: Justifying congruence by SAS

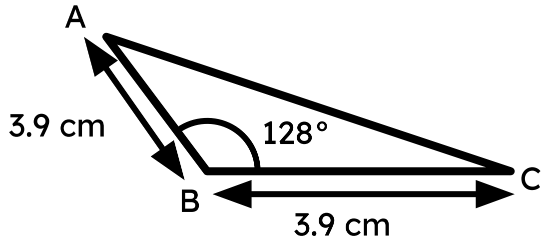
1) Construct the following triangles:

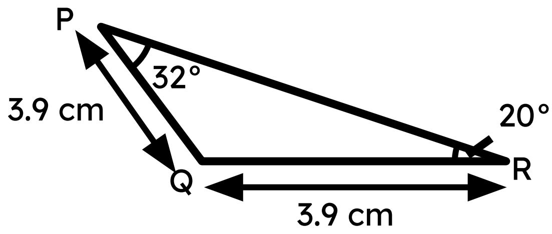
a) Triangle ABC, where AB = 6 cm, BC = 9 cm and $\angle ABC = 42°$

b) Triangle PQR, where PQ = 6 cm, QR = 9 cm and $\angle QPR = 42°$

2) Measure the edges and angles in the two triangles constructed in question 1. Are they congruent triangles?

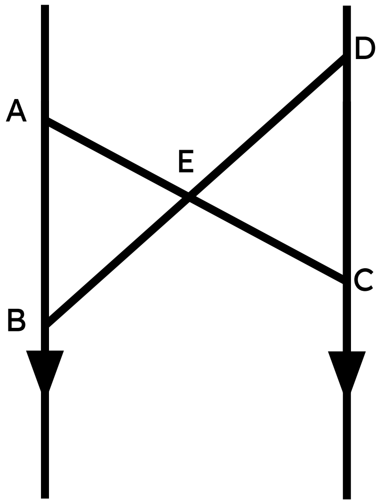
3) Prove triangle ABC is congruent to triangle PQR by SAS.





4) Given a parallelogram ABCD, prove that triangle ABC is congruent to triangle ACD by SAS.

5) Point E is the midpoint of AC and BD. Prove that triangle AEB is congruent to triangle ECD by SAS.



Task B: Two sides and an angle

1) Which of the following triangles are congruent to the triangle XYZ.